

Definition-Guided Learning and Root-Stratified Evaluation for Arabic Lexical Semantics

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Abstract

We introduce Definition-Guided Learning (DGL), a contrastive training framework that uses dictionary definitions as semantic queries and lemmas as retrieval targets, instantiated here for Arabic reverse dictionary retrieval. We construct a root-stratified Arabic lexical benchmark that limits morphological root overlap between training and test partitions. Zero-shot evaluation suggests a notable gap, with the best pre-trained model achieving only 11.16% Top@1 accuracy on this benchmark. Fine-tuning with DGL yields 41.13% Top@1 and NDCG@10 of 49.32%, an improvement of roughly 3.7 times over the strongest zero-shot baseline we tested. Our results suggest that in-batch negative sampling tends to outperform Triplet Loss in this setting, and that definition length and lexical difficulty appear to trade off in importance across retrieval depth. These findings offer preliminary evidence that lexicographic resources can serve as a useful supervisory signal for Arabic embedding learning, at least within the scope of the benchmark studied here.

1 Introduction

Recent embedding models have improved semantic similarity and retrieval performance across several NLP tasks. However, strong performance on general benchmarks does not necessarily imply robust lexical semantic representations, particularly for tasks that require fine-grained distinctions between closely related meanings (Mahajan et al., 2025). This may be particularly relevant for Arabic, where rich morphology, root-based derivation, and lexical polysemy can make identifying a lexical entry from its definition a task that involves distinguishing subtle semantic nuances, related derivations, and overlapping senses (Habash, 2010; Nehar et al., 2016). Although several Arabic-specific and multilingual embedding models are available, many are optimised primarily on usage-based training

signals rather than explicitly grounded in lexicographic semantics, where meaning is expressed through structured definitions rather than inferred from context. Progress in this direction is also limited to some extent by the relatively small number of benchmark datasets tailored to Arabic lexical resources (Alayba, 2025). We therefore construct an Arabic lexical dataset and a root-stratified benchmark, in which lemmas sharing the same morphological root are assigned to either the training or test partition, with the aim of reducing the extent to which models can rely on surface overlap and allowing some assessment of generalisation beyond morphological families. Using this benchmark, we begin with a zero-shot evaluation to get a sense of the gap. We test a set of commonly used models spanning Arabic-specific embeddings (Matryoshka and GATE (Nacar et al., 2025)), multilingual embeddings (E5 (Wang et al., 2024), Jina (Jina AI, 2024)), an Arabic lexicon-oriented model (ArabGlossBERT (Al-Hajj and Jarrar, 2021)), and a closed-source model (OpenAI (OpenAI, 2024)). None of the models we tested exceeds 11.16% Top@1 accuracy without fine-tuning, which points to a mismatch, at least on this benchmark, between off-the-shelf distributional representations and lexicographic semantics. We explore addressing this through Definition-Guided Learning (DGL), a contrastive framework that uses structured dictionary definitions as supervision for learning Arabic lexical representations. Our main contributions are: (1) an Arabic lexical dataset and a root-stratified benchmark for lexical semantic retrieval; (2) DGL, which improves Top@1 accuracy from 11.16% to 41.13% on this benchmark, roughly a 3.7-fold gain over the strongest zero-shot dense embedding baseline we evaluated; (3) an exploratory analysis of linguistic factors including definition length, polysemy, and lexical difficulty; and (4) a structured human evaluation that attempts to account for semantic ambiguity in Arabic lexical semantics.

2 Related Work

Prior work has explored lexical glosses for Arabic semantic tasks, particularly Word Sense Disambiguation (WSD). (El-Razzaz et al., 2021) introduced a BERT-based gloss WSD model achieving 89% F1, outperforming knowledge-based systems and contributing a context gloss benchmark. Similarly, ArabGlossBERT (Al-Hajj and Jarrar, 2021) fine-tunes BERT on 167,000 context gloss pairs from Arabic lexical resources, framing WSD as a binary classification task and achieving 84% accuracy. These works show that glosses are a strong supervisory signal for semantic modelling. Most directly related is the reverse dictionary task retrieving a word from its definition introduced for Arabic through KSAA-RD (Al-Matham et al., 2023). Crucially, this task was framed as embedding prediction: given a gloss, systems regress the target word’s pre-computed vector (SGNS or ELECTRA) and rank candidates by similarity to the predicted vector. The top-performing Rosetta Stone (El-Bakry et al., 2023) team followed this paradigm with an ensemble of CAMELBERT, MARBERTv2, AraBERTv2, and ELECTRA models predicting the target embedding, while Qamosy (Sibae et al., 2023) used SBERT with a Simi-Decoder. The task was later extended in KSAA-CAD (Alshammari et al., 2024), combining reverse dictionary retrieval with WSD, where ASOS (Sibae et al., 2024) achieved the best results using AraBERTv2 and AraELECTRA in a semi-encoder architecture. Across both iterations, however, the embedding-prediction framing left absolute performance low. These efforts rely on Arabic lexical resources dictionaries assembled for lookup and embedding prediction whose train/test partitions are drawn, to our knowledge, without explicit control for morphological structure. Because Arabic lemmas sharing a root remain morphologically and semantically entangled (Boudelaa et al., 2026), such partitions allow root-related lemmas to appear in both training and test, letting models exploit surface overlap rather than learn definitional meaning. We depart from these resources by releasing a root-stratified benchmark in which all lemmas of a given root are confined to a single partition, enabling a more rigorous test of generalisation beyond morphological families. Our method differs fundamentally from prior systems. Rather than regressing pre-computed target vectors, we adopt a contrastive retrieval framework (Definition-Guided Learning)

in which definitions act directly as training signals with in-batch negatives. This avoids dependence on fixed target embeddings and explicit annotations, while substantially improving over zero-shot baselines.

3 Dataset

We build our experiments on a large-scale, LMF-compliant lexicographic resource for Contemporary Arabic¹. Our working dataset comprises 90,678 (definition, lemma) pairs derived from the dictionary entries available at the time of this study. To ensure rigorous evaluation, we adopt a root-stratified split strategy in which every lemma is mapped to its morphological root, using the root field associated with each entry in the source lexicon: all lemmas belonging to a given root are assigned exclusively to either the training/development pool or the test partition, such that no root appears in both splits. The development set is drawn from the training pool via a random 10% held-out split, sharing the same root distribution as the training set; roots are disjoint only between the combined training/development pool and the test partition. This design prevents root overlap between the training/development pool and the test partition, and evaluates out-of-root generalisation rather than interpolation within seen root families. Concretely, the corpus is partitioned into 68,400 training, 7,601 development, and 14,677 test pairs, spanning 62,664 unique lemmas drawn from 6,472 distinct morphological roots; by construction, the 5,177 roots in the training pool and the 1,295 roots in the test set are disjoint. Table 1 summarises these statistics. At test time, each definitional query is ranked against the full pool of 14,677 candidate lemmas from the held-out test partition.

3.1 Data Preprocessing

All glosses and words pass through a standard Arabic normalisation pipeline before encoding: we

¹Access to the underlying lexical data is provided through the Siwar platform APIs. Researchers and developers may request API access through the Siwar developer portal at <https://siwar.ksaa.gov.sa/developers>. The record identifiers used in this study, together with the preprocessing, training, and evaluation code required to reproduce our experimental setup, are available at <https://github.com/ksaa-nlp/DGL-for-Arabic-Lexical-Semantics>. Terms of use and licensing details for accessing the Riyadh Dictionary data are published on the Siwar developer portal. As a general-purpose lexicographic resource, the dictionary does not contain personal or user-generated content, and no additional anonymisation was required.

Table 1: Dataset statistics for the root-stratified split. The development set is a random 10% hold-out of the training pool and shares its roots.

	Train	Dev	Test
Pairs	68,400	7,601	14,677
Unique lemmas	53,297	–	9,367
Unique roots	5,177	–	1,295
Total: 90,678 pairs, 62,664 lemmas, 6,472 roots			

remove diacritics and Quranic annotation marks, unify alef variants (أ, إ, ئ) to the bare form (ا), and normalise ta marbuta (ة) to ha (ه), reducing orthographic sparsity and improving cross-entry consistency. Each normalised gloss is then formatted as a retrieval query using the template “(ما الكلمة التي تعني): definition”, which conditions the encoder to treat the input as a definitional query rather than a standalone sentence. This query serves as the anchor in our contrastive framework, with the corresponding lemma as the retrieval target.

4 Methodology

4.1 Contrastive Learning

Contrastive learning has emerged as a dominant paradigm in representation learning, demonstrating strong performance across both vision and language domains. In its original formulation, the framework trains an encoder to produce representations that are similar for semantically related inputs (positive pairs) and dissimilar for unrelated ones (negative pairs). This objective was formalised in self-supervised settings by (Chen et al., 2020) for vision, and subsequently adapted to natural language by (Gao et al., 2021), which showed that simple in-batch contrastive objectives yield high quality sentence embeddings without requiring explicit supervision. In this work, we adopt contrastive learning as the core training objective for the definition-to-word retrieval task. Formally, given a set of training instances (anchor, positive), where anchor is a definitional query and positive is its corresponding target lemma (positive), the model is trained to minimise the distance between anchor and positive representations while maximising the distance to all other representations in the embedding space. We explore two contrastive training strategies, Triplet Loss and In-Batch Negative Sampling which, differ in how negative examples are constructed, as described in the following sub-

sections. **Triplet Loss with Hard Negatives** Inspired by (Nacar et al., 2025), who demonstrated the effectiveness of Triplet-based contrastive objectives for Arabic word embedding, we adopt Triplet Loss as our first training strategy. This objective operates on explicitly constructed triplets of the form (anchor, positive, negative): In our formulation:

- **Anchor (a):** a definitional query constructed from the lexical gloss, formatted as: “(ما الكلمة التي تعني): definition”
- **Positive (p):** the target Arabic lemma corresponding to the given definition.
- **Negative (n):** a semantically related but incorrect lemma, selected using a pre-trained embedding model.

Negative samples are generated using E5-large (Wang et al., 2024) by retrieving the top-five most similar lemmas for each anchor, excluding the gold positive. This strategy produces hard negatives semantically proximate candidates that are nevertheless incorrect which are known to yield stronger training signal than random negatives. **In-Batch Negative Sampling** The second contrastive strategy we employ is In-Batch Negative Sampling (Gao et al., 2021), which avoids the need for explicit negative construction by treating all other positive samples within the same training batch as negatives for a given anchor.

- **Anchor (a):** a definitional query constructed from the lexical gloss, formatted as: “(ما الكلمة التي تعني): definition”
- **Positive (p):** the target Arabic lemma corresponding to the given definition.

Concretely, given a batch of N anchor–positive pairs, the loss for each anchor is computed as a SoftMax over cosine similarity to its one positive representation and $N-1$ negative representations in the batch, scaled by a temperature parameter τ . This formulation corresponds to the `MultipleNegativesRankingLoss` implemented in the `sentence-transformers` library (Reimers and Gurevych, 2019). The key advantage of this approach is that each training step generates $N-1$ implicit negative per anchor, providing a dense and diverse contrastive signal that scales naturally with batch size. In our experiments, we use a batch size

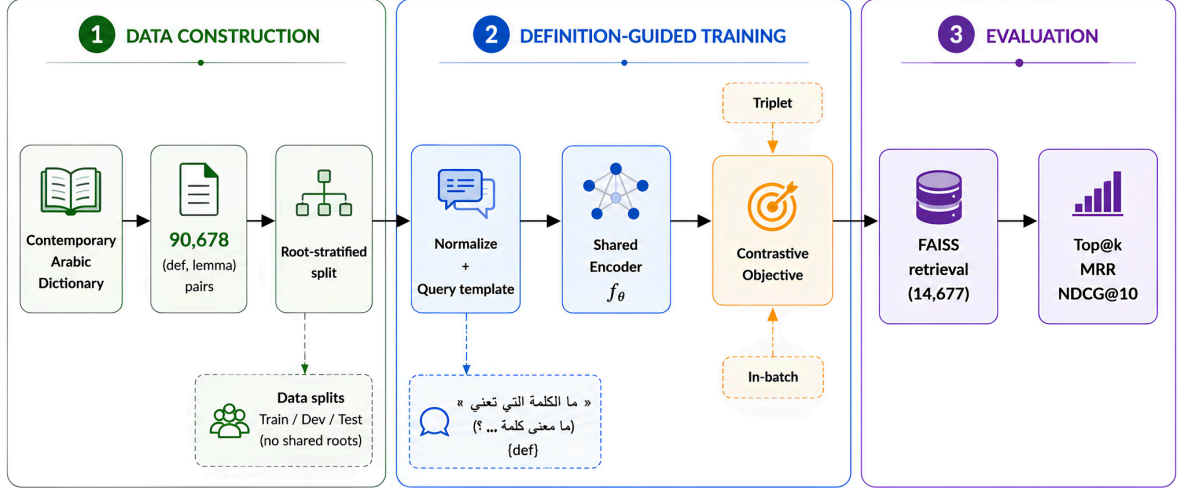


Figure 1: Overview of the proposed Definition-Guided Learning (DGL) method. The development set is sampled from the training pool and shares its root distribution; root disjointness applies only to the training/development pool versus the test partition (§3).

of 256, yielding 255 implicit negatives per instance substantially richer than the single hard negative used in the Triplet setting.

4.2 Models

We fine-tune the three strongest *open-weight* embedding models from the zero-shot evaluation (Section 5.1). The top-ranked proprietary model (OpenAI text-embedding-3-small) is excluded because it cannot be fine-tuned in our setting. The selected models are: **Matryoshka**, an Arabic Matryoshka representation-learning model (Nacar et al., 2025); **GATE**; and **E5-large** (Wang et al., 2024). All three are fine-tuned under both the Triplet Loss and In-Batch Negative Sampling objectives using identical hyperparameters to ensure a fair comparison.

4.3 Training Configuration

All experiments were conducted using a batch size of 256, providing 255 in-batch negatives per anchor, on a single NVIDIA A100 GPU. To ensure a controlled comparison across models and training objectives, all configurations were trained for five epochs using a learning rate of $5e-5$ (the default value for Sentence Transformer TrainingArguments in the HuggingFace Transformers library, not explicitly overridden). Optimisation was performed using AdamW (Loshchilov and Hutter, 2019) with a linear warmup ratio of 0.1. Training employed BF16 mixed precision, and for the Matryoshka

model, the embedding dimensionality was set to 768. A fixed random seed of 42 was applied consistently across random, numpy, and torch for reproducibility; reported results correspond to a single run per configuration (single seed). Fine-tuning used sentence-transformers version 5.6.0. The Matryoshka checkpoint corresponds to the HuggingFace repository Omartificial-Intelligence-Space/Arabic-Triplet-Matryoshka-V2(commit b581354), fine-tuned from aubmindlab/bert-base-arabertv02 (AraBERT v0.2; 135,193,344 parameters); we cite this specific checkpoint via (Nacar et al., 2025), distinguishing it from the general Matryoshka Representation Learning method (Kusupati et al., 2022). At inference, queries and candidate lemmas are encoded with the fine-tuned model and retrieved with FAISS (Johnson et al., 2019) using inner-product (cosine) similarity over the candidate pool of 14,677 test-partition lemmas.

5 Results

The evaluation is conducted using standard information retrieval metrics, namely **Top@k**, **Mean Reciprocal Rank (MRR)**, and **Normalized Discounted Cumulative Gain (NDCG@10)**. Top@k measures the proportion of queries for which the correct lemma appears within the **Top@k** retrieved candidates, capturing retrieval performance at different cut-off levels. **MRR** evaluates the average reciprocal rank of the first correct answer, reflecting how early the correct result appears in the

ranked list. **NDCG@10** assesses ranking quality by assigning higher importance to correct answers appearing at top positions, making it particularly sensitive to the ordering of retrieved candidates.

5.1 Zero-Shot Baseline Results

We first evaluate six pre-trained embedding models together with a BM25 lexical baseline (Robertson and Zaragoza, 2009) in a zero-shot setting, without any task specific fine-tuning, to establish the performance ceiling of off the shelf retrieval on Arabic definition-to-word matching. Besides the three models carried forward to fine-tuning (Matryoshka, GATE, and E5-large; Section 4.2), the pool includes the proprietary OpenAI text-embedding-3-small, jina-embeddings-v5-text-nano (Jina AI, 2024), and ArabGlossBERT (Al-Hajj and Jarrar, 2021). Results are shown in Table 2.

We observe that performance is uniformly low across all models and metrics. The strongest embedding model, Matryoshka, achieves a Top@1 of 11.16% and an NDCG@10 of 24.45%. The BM25 lexical baseline attains a closely comparable Top@1 (10.13%), reflecting cases where a definition shares surface tokens with its target lemma; however, BM25 is clearly surpassed by Matryoshka at every deeper cut-off (e.g. Top@10: 36.75% vs. 12.99%) and on ranking quality (NDCG@10: 24.45% vs. 11.96%). This suggests that even the best-performing general-purpose embedding model is unable to reliably retrieve the correct lemma, confirming a substantial performance gap between pre-trained representations and the requirements of this task. These results indicate that Arabic definition-to-word retrieval cannot be adequately addressed by zero-shot application of existing embedding models, regardless of their general-purpose capabilities. Notably, OpenAI text-embedding-3-small, a state-of-the-art proprietary model, ranks fifth among the evaluated embedding models (Top@1: 4.26%), behind Matryoshka, GATE, E5-large, and Jina, suggesting that scale alone does not compensate for the absence of task-specific training on Arabic lexical data.

5.2 Fine-Tuning Results: Triplet Loss

Following the zero-shot evaluation, we fine-tune the top three models Matryoshka, GATE, and E5-large under the Triplet Loss objective with hard negatives generated by E5-large. Results are re-

ported in Table 3. Fine-tuning with Triplet Loss yields a substantial improvement over the zero-shot baseline for most models and metrics, most notably Top@1 and MRR; for E5-large, however, deeper-cutoff metrics do not improve (Top@10: 26.68% zero-shot vs. 24.75% with Triplet Loss), indicating that the gains are not uniform across ranking depth. Matryoshka achieves a Top@1 of 30.21% and an NDCG@10 of 35.09%, a relative gain of roughly $2.7\times$ over its zero-shot counterpart (11.16%). These results demonstrate that task-specific contrastive training with explicit hard negatives provides a strong inductive signal for Arabic lexical retrieval. We observe that all three models converge to similar performance levels under this objective, with Matryoshka and GATE separated by less than 0.3 percentage points in Top@1 (30.21% vs. 29.96%). This convergence suggests that the Triplet Loss objective imposes a training ceiling that is not strongly dependent on model architecture, potentially due to the limited diversity of hard negatives introduced by a fixed negative sampling strategy.

Fine-Tuning Results with Triplet Loss: Impact of Refined Hard Negatives To improve the construction of hard negative samples for Arabic semantic retrieval, we introduce a structured multi-stage pipeline that integrates neural retrieval, large language model (LLM) based selection, and linguistically informed filtering. We begin by leveraging pretrained embedding models to retrieve a pool of semantically similar candidates for each query-positive pair. This retrieval step ensures that candidate negatives lie in a dense semantic neighborhood, increasing the likelihood of generating challenging and informative contrasts. To refine candidate selection, we employ a GPT-4 based model as a semantic adjudicator guided by a carefully designed prompt enforcing strict lexical and semantic constraints. These include excluding synonyms, paraphrases, and semantically entailed alternatives, while encouraging the selection of contextually plausible yet incorrect candidates. To further mitigate false negatives, we incorporate multiple linguistically informed filtering layers: synonym filtering using external lexical resources, derivative filtering to remove morphologically related forms, and definition-level similarity filtering based on embedding similarity between lexical definitions. To assess the impact of this refinement, we conducted a manual qualitative review of a random sample of 50 hard negatives,

Table 2: Zero-shot retrieval results. Across diverse model families, no system exceeds 11.16% Top@1. The strongest embedding model, Matryoshka, leads across all six metrics, including a narrow margin over the lexical BM25 baseline on Top@1 (11.16% vs. 10.13%), underscoring the persistent gap between off-the-shelf representations and the requirements of lexicographic definition-to-word retrieval.

Category	Model	Top@1	Top@3	Top@5	Top@10	MRR	NDCG@10
Sparse	BM25	10.13%	11.95%	12.62%	12.99%	11.14%	11.96%
Arabic-specific	Matryoshka	11.16%	24.40%	29.79%	36.75%	19.15%	24.45%
	GATE	10.21%	23.79%	29.03%	35.42%	18.23%	23.45%
Multilingual	E5-large	9.74%	15.42%	19.84%	26.68%	14.00%	17.77%
	Jina	4.69%	10.09%	14.28%	20.00%	8.62%	12.00%
Closed-source	OpenAI	4.26%	8.18%	10.91%	15.61%	7.17%	9.65%
Lexicon-oriented	ArabGlossBERT	2.76%	4.86%	6.37%	8.65%	4.29%	5.61%

Table 3: Fine-Tuning Results: Triplet Loss with Hard Negatives vs. Refined Hard Negatives.

Setting	Model	Top@1	Top@3	Top@5	Top@10	MRR	NDCG@10
Triplet Loss	Matryoshka	30.21%	34.94%	36.93%	39.66%	33.14%	35.09%
	GATE	29.96%	34.76%	36.67%	39.80%	32.95%	35.00%
	E5-large	17.97%	21.25%	22.77%	24.75%	20.04%	21.37%
Refined Hard Negatives	Matryoshka	16.26%	30.12%	31.66%	33.01%	23.32%	25.76%
	GATE	15.14%	30.04%	31.71%	33.03%	22.73%	25.32%
	E5-large	17.55%	26.23%	28.75%	31.03%	22.27%	24.41%

covering both retrieved-candidate and fallback-generated cases; the selected negatives were qualitatively judged, in this limited manual sample, to be higher quality i.e., semantically close but clearly incorrect compared to negatives from our earlier, pre-refinement pipeline. We evaluate the impact of these refined hard negatives on downstream retrieval performance, and the results are presented in Table 3. Despite the improved linguistic quality of the constructed negatives confirmed by this manual review, the performance gains remain limited. As shown in Table 3, the results indicate that enhancing the precision and linguistic validity of hard negatives does not necessarily lead to improved retrieval performance. Refined hard negatives substantially reduce performance for Matryoshka and GATE. For E5-large, Top@1 decreases slightly, while the deeper-cutoff and ranking metrics improve. We had initially hypothesised that this may stem from a reduction in the diversity of training signals introduced by our filtering pipeline. However, our comparison only contrasts the unfiltered and fully filtered endpoints, and cannot distinguish gradual diversity loss across filtering steps from the effect of a single dominant filtering component (e.g., retrieval, synonym filtering, root-based filtering, or GPT-4 adjudication). We therefore do not treat the diversity-based explanation as an es-

tablished finding, and instead report the observed performance drop as an unexplained empirical observation; a step-wise ablation isolating each filtering component is left for future work. In contrast, in-batch negatives, despite being noisier and less controlled, achieve stronger and more robust performance in our experiments (Table 4), which we similarly report as an empirical observation without a fully verified causal explanation.

5.3 Fine-Tuning Results: In-Batch Negative Sampling

We next evaluate the same three models fine-tuned under the In-Batch Negative Sampling objective. Results are reported in Table 4. In-Batch Negative Sampling yields the strongest results across all configurations. Matryoshka achieves a Top@1 of 41.13% and an NDCG@10 of 49.32%, a roughly 3.7-fold improvement over the zero-shot baseline and a clear gain over the Triplet Loss configuration. These results establish in-batch negatives as the preferred training objective for this task. We observe a consistent and clear advantage of In-Batch Negatives over Triplet Loss across all three models and all six evaluation metrics. The improvement is particularly pronounced in Top@1 (+10.92% points for Matryoshka) and NDCG@10 (+14.23% points), suggesting that denser negative

signals not only improve exact retrieval precision but also enhance the overall ranking quality of retrieved candidates. We additionally report a hybrid configuration (Matryoshka + BM25) that linearly combines the fine-tuned dense scores with lexical BM25 scores over the same candidate pool. BM25 scores were computed using the `rank_bm25` Python library (BM25Okapi) with the standard default parameters $k_1 = 1.5$ and $b = 0.75$, adopted as-is without tuning on our data; Arabic text was tokenised via the same diacritic-removal and letter-normalisation preprocessing used elsewhere in our pipeline, followed by whitespace tokenisation. Both score sets are normalised and fused as $\alpha s_{\text{dense}} + (1 - \alpha) s_{\text{BM25}}$ with $\alpha = 0.7$, a fixed heuristic weighting favouring the dense signal and not tuned on a held-out development set. Despite augmenting dense retrieval with an explicit lexical signal, the hybrid underperforms the purely dense Matryoshka model (Top@1: 32.64% vs. 41.13%), indicating that after fine-tuning the dense representations already subsume the lexical overlap BM25 captures, while BM25’s noisier signal dilutes the ranking. All results reported above use the test-only candidate pool (14,677 lemmas, duplicates retained) at true batch size 256. Full sensitivity results across batch sizes and candidate-pool constructions are reported in Table 6 (Appendix E).

6 Linguistic Feature Analysis

To better understand model behaviour, we examine how retrieval performance relates to three linguistic properties: definition length (the token count of the normalised gloss), word sense count, and lexical difficulty (a 1–5 score assigned by GPT-5). Using the predictions of our selected Matryoshka configuration (fine-tuned with In-Batch Negatives), we train a Random Forest Regressor to predict each retrieval metric from these features and report the resulting importance scores across all six metrics (Figure 4). Definition length is the most informative feature for exact retrieval, with the highest importance at Top@1 (44.3%), though its weight declines as k increases (35.5% at Top@10). Lexical difficulty shows the opposite trend: it is least important for Top@1 (27.4%) but rises steadily with k , overtaking definition length to become the dominant feature at Top@10 (36.4%) and remaining strong for NDCG@10 (33.9%). This crossover is informative: definition length largely governs whether the correct word is retrieved at the very

top, whereas lexical difficulty more strongly shapes performance over the wider candidate ranking captured by Top@10 and NDCG@10. Word sense count contributes least and most uniformly across metrics (26.9% - 29.6%), suggesting that, within our evaluation setting, polysemy has a comparatively minor effect relative to the other two features.

7 Discussion

The results presented above provide strong evidence that lexical definitions constitute a rich and effective supervisory signal for training semantically grounded Arabic embedding models. The roughly 3.7-fold improvement in Top@1 accuracy from 11.16% in the zero-shot setting to 41.13% under our selected fine-tuned configuration demonstrates that the semantic content encoded in dictionary glosses is sufficiently structured and discriminative to guide contrastive learning toward meaningful lexical representations. This improvement, however, is not uniform across training strategies, and the choice of contrastive objective proves decisive. The consistent superiority of In-Batch Negative Sampling over Triplet Loss across all three models and all six evaluation metrics warrants careful interpretation. The performance gap, ranging from +10.92 to +17.03 Top@1 percentage points, is not marginal. We attribute it primarily to a difference in the quantity and diversity of the contrastive signal available during training, though we note that these two factors are entangled in our comparison and cannot be fully separated. While Triplet Loss restricts each training instance to a single hard negative, In-Batch Negative Sampling exposes the model to 255 implicit negatives per instance at a batch size of 256, drawn from diverse semantic domains within the same batch. This density and diversity of negative signals plausibly enables the model to learn finer-grained distinctions across the lexical space, not just between an anchor and one hard negative, but across a broad neighborhood of semantically proximate lemmas simultaneously. Notably, this runs somewhat counter to the standard hard-negative mining intuition that harder, semantically closer negatives are most informative: in our setting, the broader and noisier in-batch signal outperformed a single curated hard negative, which makes the advantage of diversity a non-trivial finding rather than a foregone conclusion. This finding carries a practical implication

Table 4: Fine-tuning results with In-Batch Negative Sampling. Matryoshka + BM25 is the hybrid dense lexical configuration ($\alpha = 0.7$).

Model	Top@1	Top@3	Top@5	Top@10	MRR	NDCG@10
Matryoshka	41.13%	47.67%	51.93%	57.40%	45.65%	49.32%
GATE	41.28%	47.76%	51.86%	57.27%	45.73%	49.33%
E5-large	35.00%	40.66%	43.77%	48.21%	38.72%	41.65%
Matryoshka + BM25	32.64%	40.88%	45.02%	50.41%	37.86%	40.85%

for future work: since the quality of in-batch negatives is directly tied to batch composition, batches containing semantically diverse lemmas are likely to generate more informative implicit negatives than batches dominated by lemmas from the same semantic domain or root family, suggesting that diversity-aware batch construction strategies could yield further gains.

8 Error Analysis

We analyse 100 test instances for which the fine-tuned retrieval model failed to recover the correct lemma within the Top@10 retrieved candidates (Top@1 = Top@3 = Top@5 = Top@10 = False for all cases). Manual analysis of the Top@1 prediction score for each failure case reveals that these errors are not uniform in nature, but instead fall along a continuum of residual semantic relatedness. At the high end of this continuum (score $\geq 65\%$), the model retrieves a lemma that is a near-perfect semantic synonym of the gold lemma, frequently a case where the dictionary’s own definition of the gold lemma consists primarily of the predicted word itself, as in ((انسكاب)) (اصطباب) being predicted as ((انسكاب)) with a score of 92%. These are arguably not retrieval failures in a strict semantic sense, but rather artefacts of an evaluation protocol that recognises only a single gold lemma per definition despite genuine synonymy in the lexicon; in several of these cases the predicted and gold lemmas additionally share the same morphological pattern (wazn), which a strict exact-match evaluation penalises even though both forms are valid and semantically interchangeable. At the intermediate range (score 45%–65%), the model retrieves a lemma from the same semantic field as the gold lemma typically a hypernym, a co-hyponym, or a component concept embedded within the gold definition suggesting that the model falls back on the closest available semantic neighbour when it cannot resolve the fine-grained

distinction required for exact retrieval, as seen with ((طلب وضع النقط في كتاب)) (استنقاط) being predicted as ((نقط)). At the lowest end of the continuum (score < 45), the model’s top prediction bears little discernible semantic relationship to the gold lemma; these cases most closely resemble genuine retrieval failures and are concentrated among definitions describing rare, culturally specific, or highly compositional concepts such as ((صمغريات)), a specialised term for a breed of war camels that is likely underrepresented in the training distribution. Representative examples from each band are presented in Table 5. Taken together, this gradient illustrates a key limitation of strict automatic evaluation. Near-Synonym and Related-Concept cases make up most of the failures in this 100-sample analysis, while Weak-Match cases are the minority. Most of what automatic metrics count as “wrong” are thus near misses valid synonymy or morphological near identity a human would accept rather than true breakdowns in semantic association, a distinction strict Top@ k metrics cannot capture. This motivated the human evaluation in the next Section. To assess whether fine-tuning altered the nature of these residual errors, we traced the same failure cases back through the zero-shot model’s predictions. Zero-shot scores in the Near-Synonym band were already fairly high (0.69–0.84), and fine-tuning mostly pushed confidence further toward the same near-synonym rather than resolving the underlying single-gold-lemma limitation. In several cases, however, fine-tuning changed the model’s top prediction altogether: for the definition of ((النبيذ)) (كأس), the zero-shot model retrieved ((محتسبه)), whereas the fine-tuned model retrieved ((جمعه)) instead. This points to a shift in the model’s semantic reasoning rather than a simple boost in confidence. Overall, DGL appears to pull residual errors closer to genuine semantic neighbours, but it does not remove this class of near-miss errors entirely.

9 Ablation Study

To disentangle these factors, we ran a batch-size sensitivity analysis on the Matryoshka model using true batch sizes of 32, 64, 128, and 256 (up to 255 real in-batch negatives per anchor), all trained with `CachedMultipleNegativesRankingLoss` to keep the loss mechanism identical across the sweep. Results show an overall improvement, with minor non-monotonic fluctuations, across all metrics on both the full and test-only corpora as batch size increases (full-corpus NDCG@10: 40.43% $\rightarrow$ 41.47%; test-only NDCG@10: 48.35% $\rightarrow$ 49.32%, for batch sizes 32 $\rightarrow$ 256). Full results are reported in Table 6 in Appendix E.

10 Human Evaluation

To complement the automatic metrics, we conducted a production-oriented human evaluation using free-form, user-authored queries. Four PhD holders in Arabic, balanced by gender, wrote definitions in their own words without being restricted to the dictionary’s phrasing. All 160 queries were evaluated using our selected configuration (Matryoshka fine-tuned with in-batch negative sampling), retrieving the top five candidates per query. Two Arabic-language evaluators, neither of whom authored the queries, independently judged each candidate as a valid synonym or not, yielding 800 binary judgments. They agreed on 657 (82.1%; 95% query cluster bootstrap CI: 79.8–84.5; Cohen’s $\kappa = 0.629$, 95% CI: 0.576–0.679). The 143 disagreements, spanning 103 queries, were resolved by a third adjudicator (90 valid, 53 invalid), yielding a final reference of 502 valid and 298 invalid judgments. Based on this reference, micro-averaged Accuracy@1, @3, and @5 (Hit@ k) were 75.0% (120/160; 95% Wilson CI: 67.8–81.1), 88.8% (142/160; 82.9–92.8), and 93.8% (150/160; 88.9–96.6). This Accuracy@1 is not directly comparable to the 41.13% obtained on the held-out test set: the human protocol accepts any valid synonym rather than a single gold lemma, a stricter exact-match criterion that our error analysis (Appendix C) shows penalises many valid near-synonym predictions. The consistent increase from rank 1 to rank 5 nevertheless shows that the correct lemma is frequently retrieved among top candidates even when not top-ranked, suggesting the learned embedding space places correct answers close to naturally phrased queries. As a supplementary check, GPT-5.6-Sol agreed with the hu-

man reference on 602 of 800 judgments (75.3%; 95% CI: 71.0–79.3; $\kappa = 0.527$, 95% CI: 0.455–0.598), with high precision (96.6%) but lower recall (62.7%), suggesting its scores are a conservative lower bound. These results were not used to replace human judgments or compute the primary reported results.

11 Conclusion

This work introduced Definition-Guided Learning (DGL), a contrastive framework that uses dictionary definitions as a supervisory signal for Arabic semantic representation learning, evaluated through reverse dictionary retrieval on a root-stratified benchmark we contribute as a methodological resource. Fine-tuning with in-batch negative sampling raised Top@1 accuracy from an 11.16% zero-shot baseline to 41.13%, with retrieval performance shaped by definition length, polysemy, and lexical difficulty. Our error analysis and human evaluation further show that many apparent failures are in fact valid synonyms penalised by strict exact-match scoring, underscoring the value of dictionary definitions as a rich, under-exploited resource for Arabic NLP and motivating future work on relaxed evaluation protocols and broader lexicographic applications beyond reverse dictionary retrieval.

Limitations

Our evaluation relies on a single MSA lexicographic resource, and we cannot yet confirm that our approach generalises to other Arabic dictionaries, which may differ in root-family distributions or lexical coverage, or to dialectal Arabic. Lexical difficulty scores were assigned using GPT-5 rather than human linguists, which may introduce systematic biases not present in fully human-annotated difficulty judgements. Our human evaluation, while informative, involved a limited pool of four query authors and a set of 160 free-form queries, which may not fully capture the diversity of real-world user phrasing; while inter-annotator agreement on the candidate-level validity judgments was substantial (Cohen’s $\kappa = 0.629$ before adjudication), each query itself was authored by a single annotator, so query-generation biases specific to this small pool cannot be fully ruled out. All annotators involved in the human evaluation (query authors, candidate evaluators, and the adjudicator) were volunteers who received no monetary compensation for their

participation. No formal institutional ethics board (IRB) approval was sought, as the study involved no collection of sensitive personal information, no vulnerable populations, and no risk beyond that of routine linguistic annotation. Finally, our framework was evaluated solely on reverse dictionary retrieval; while we motivate Definition-Guided Learning as a general approach to semantic representation learning, its effectiveness on other downstream tasks, such as word sense disambiguation or definition generation, has not been empirically verified in this work.

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A Lexical Difficulty Annotation Prompt

To assign a difficulty score to each lemma in our dataset, we queried GPT-5, accessed using the structured prompt shown in Figure 2. Words were submitted in batches of 10 per call, in a fixed order, and the model was instructed to return exactly one numeric score per word in the same order, without any accompanying explanation, to enable automatic and unambiguous parsing of the output.

```
Task: Assign a difficulty level (1-5) to each Arabic word in Modern Standard Arabic.
Consider ALL factors:
- Frequency
- Age of acquisition
- Concreteness
- Word length
- Morphological complexity
- Semantic transparency

Scale:
1 Very easy
2 Easy
3 Moderate
4 Difficult
5 Very difficult

Strict rules:
Return ONLY numbers
One number per line
Same order
No explanations
```

Figure 2: Structured prompt used to assign lexical difficulty scores via GPT-5.

Example. Given the input words (طاولة), (عين), and (استنباط) , the model returned the sequence 2, 1, 4 corresponding, in order, to a common polysemous noun, a basic everyday object, and a morphologically complex derived form, respectively.

Validation. We manually verified the difficulty assignments on a random sample of words and compared them against human judgments.

B Hard Negative Selection Prompt

After retrieving and filtering candidate negatives as described in Section 5.2 (E5-large retrieval followed by synonym/root-based filtering), GPT-4 was prompted to select a single hard negative from the remaining candidates for each anchor definition, using the prompt shown in Figure 3.

You are an Arabic lexicography judge for hard negative mining.

Choose ONE hard but valid negative.

Rules:

- Prefer selecting from the retrieved candidates.
- The hard negative must be semantically close, but NOT a synonym and NOT a correct answer.
- Avoid words sharing the same stem/root if they preserve the same meaning.
- Avoid easy or unrelated words.
- If all candidates are weak or invalid, generate a semantically related Arabic lexical candidate. It may later be mapped back to the nearest corpus word.

Figure 3: Prompt used to select a hard negative from filtered E5-large candidates via GPT-4, for Triplet Loss training.

Example. For the anchor definition corresponding to the gold lemma (انصباب)(انسكاب), “spilling”, after filtering out the synonym (انسكاب) itself and root-sharing forms, GPT-4 selected (تسرب) (“leakage”) as a semantically related but non-synonymous and incorrect hard negative.

Fallback generation. In cases where none of the filtered candidates satisfied the difficulty and validity criteria, GPT-4 generated a new Arabic lexical candidate directly, which was subsequently mapped to its nearest match in the corpus vocabulary or discarded if no sufficiently close match existed.

C Error Analysis

Table 5: Representative failure cases across the error-severity spectrum (fine-tuned Matryoshka). Most failures are near-synonyms or related concepts that a single-gold, exact-match protocol penalises despite genuine semantic validity.

Band	Definition (query)	Gold lemma	Top@1 prediction	Score
Near-Synonym	انسكاب	اصطباب	انسكاب	0.927
	تحويل وتبديل	اقلاب	تحويل	0.812
	استهانه واستخفاف	استبساط	استخفاف	0.796
Related-Concept	طلب وضع النقط في كتاب او غيره	استنقاط	نقط	0.644
	اسرع اشتعالا وانتقادا	اورى	اشظى	0.642
	لباس مناسبه خاص كالتسويج او التوليه	خلعه	زي	0.641
Weak-Match	النبيذ	كاس	جعه	0.399
	نوع خفيف من النعال يستعمل في بلاد المغرب	بلغه	شلجم	0.381
	فصيله من الابل تهاجم العدو عند سماع صوت الرصاص	صخريات	هنوف	0.368

D Linguistic Feature Analysis

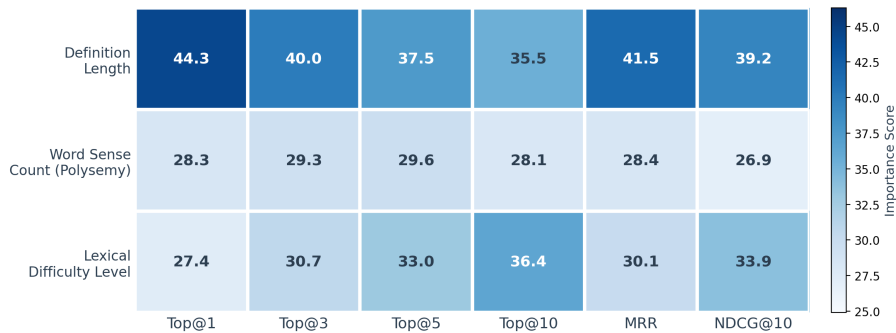


Figure 4: Feature importance heatmap across all six retrieval metrics.

E Batch-Size and Corpus Sensitivity Analysis

Table 6 reports full results across all tested batch sizes and both candidate-pool constructions, for all three fine-tuned models.

Table 6: Sensitivity of fine-tuning results to true batch size (32–256, i.e., up to 255 in-batch negatives per anchor) and candidate-pool construction (test-only, 14,677 lemmas with duplicates retained, vs. deduplicated combined train+test pool, 62,664 lemmas). All models trained with CachedMultipleNegativesRankingLoss to keep the loss mechanism identical across the sweep. Batch size 256 / test-only corresponds to the headline results in Table 4.

Batch Size	Model	Corpus	Top@1	Top@3	Top@5	Top@10	MRR	NDCG@10
32	GATE-AraBert	Test-only	40.08%	46.92%	50.83%	56.28%	44.67%	48.32%
		Full	32.58%	40.52%	44.10%	48.96%	37.54%	40.26%
	Matryoshka-V2	Test-only	40.35%	47.09%	50.71%	56.14%	44.79%	48.35%
		Full	33.14%	40.57%	44.04%	48.82%	37.82%	40.43%
	E5-large	Test-only	32.24%	37.32%	39.90%	43.59%	35.48%	37.96%
		Full	27.93%	33.37%	35.75%	38.69%	31.27%	33.04%
64	GATE-AraBert	Test-only	40.60%	47.60%	51.72%	56.94%	45.21%	48.92%
		Full	33.02%	40.96%	44.58%	49.79%	38.03%	40.82%
	Matryoshka-V2	Test-only	40.74%	47.61%	51.73%	57.06%	45.31%	49.01%
		Full	33.10%	41.02%	44.79%	49.80%	38.13%	40.91%
	E5-large	Test-only	32.81%	38.16%	41.19%	45.02%	36.34%	39.01%
		Full	28.81%	33.98%	36.75%	40.16%	32.18%	34.07%
128	GATE-AraBert	Test-only	41.02%	47.85%	51.90%	57.12%	45.57%	49.21%
		Full	33.71%	41.13%	44.83%	49.91%	38.46%	41.18%
	Matryoshka-V2	Test-only	41.12%	47.79%	52.01%	57.25%	45.64%	49.27%
		Full	33.66%	41.21%	44.85%	50.00%	38.49%	41.23%
	E5-large	Test-only	34.20%	39.72%	42.87%	47.03%	37.86%	40.69%
		Full	29.40%	35.54%	38.24%	41.95%	33.18%	35.27%
256	GATE-AraBert	Test-only	41.28%	47.76%	51.86%	57.27%	45.73%	49.33%
		Full	33.81%	41.58%	45.31%	50.30%	38.71%	41.47%
	Matryoshka-V2	Test-only	41.13%	47.67%	51.93%	57.40%	45.65%	49.32%
		Full	33.86%	41.53%	45.36%	50.28%	38.73%	41.47%
	E5-large	Test-only	35.00%	40.66%	43.77%	48.21%	38.72%	41.65%
		Full	29.86%	36.10%	39.20%	42.99%	33.81%	36.00%